

Assumption (thought)	P (prior)	P (after)	KLD
healthconscious(mary):	0.50	0.82228	0.0735
wantsNotDrink(mary,coffee):	0.40	0.9208	0.1873
wantsNotDrink(mary,coke):	0.50	0.92	0.1151
wantsNotDrink(mary,fruitTea):	0.60	0.0199	1.4636
wantsNotDrink(mary,peppermintTea):	0.60	0.4	0.0432
wantsNotDrink(mary,proteinShake):	0.80	0.9831	0.0182
wantsNotDrink(mary,redBull):	0.80	0.98	0.0176
wantsNotDrink(mary,tea):	0.50	0.88	0.0973
wantsNotDrink(mary,water):	0.50	0.1	0.4047
			2.4207

5.4.6 Comprehension modelling as optimization model

The preceding discussion shows that we can conceptualize the comprehension of implicatures in the form of an optimization model having the following structure:

1. Eliminate interpretive hypotheses from consideration that do not satisfy all relevance expectations (such as Int 2).
2. If interpretive hypotheses have the same KL measure, consider only the one with the smallest Herbrand base.
3. From among the remaining interpretive hypotheses, accept the one with the highest KL measure.

This optimization model²¹ allows us to construct quantitative predictions about the interpretation that the relevance-theoretic comprehension heuristic would pick out. These quantitative predictions could be used in experiments investigating the heuristic nature of relevance-guided comprehension.

6 Conclusion

In this paper we have developed an *as-if*-optimization model of implicature comprehension that captures the insights of relevance theory in probabilistic terms and implemented a computational model of it in the probabilistic programming language ProbLog. This model can serve as a tool to investigate the heuristic nature of the relevance-theoretic comprehension procedure by way of comparing experimental results with quantitative model predictions.

The model shares with classical relevance theory its focus on the interpretive hypothesis-evaluation part of non-demonstrative inference processes, assuming that the hypothesis-generation part of non-demonstrative inference is inamenable to explanatory study Sperber and Wilson (1995, p. 68). In consequence, we require candidate interpretive hypotheses of utterances to be manually constructed by the experimenter/s/programmers.

Moreover, this model focusses on the audience's part in communication, the comprehension of utterances, as does classical relevance theory. The reason for this is

²¹'*as-if*-optimization' in the terms of Gigerenzer (2004, p. 64)

that given the relevance-seeking nature of human minds, speakers must take this nature into account if they want to be understood and choose their utterances such that the audience only needs to follow where their relevance expectation-satisfaction search will lead them. This means that responsibilities, and sophistication in cognitive processes, are unevenly distributed, making the comprehension phase of utterance interpretation more amenable to study. This does not mean, however, that the speaker's perspective is not important to study; on the contrary, extending a relevance-theoretic account of communication to the speaker's perspective is a strongly desirable goal. For this goal to be reached, it is important to gain a better understanding of the inference processes involved in metacognition: how to cognizers (and in particular, communicators) estimate what expatations of relevance others have?

Other areas where our model needs extension include the modelling of higher-level explicature comprehension (speech act and propositional attitude recognition, modal meanings) as well as to instances of metarepresentational utterance use, as for example in irony, direct and indirect quotation, free indirect speech and thought reports. This will be left to future research.

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